

Lan E. Luo

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Curriculum Vitae

EDUCATION

Columbia University
Ph.D. Marketing
M.Phil. Marketing

New York, NY
2026 (Expected)
2024

Yale University
B.A. Economics; B.A. Statistics & Data Science (Honors)

New Haven, CT
2020

RESEARCH INTERESTS

Substantive: Data-Driven Design, User-Generated Content, Digital Advertising, Social Issues

Methodological: Deep Generative Modeling, Interpretable Machine Learning, Applied Econometrics, Causal Inference, Probabilistic Machine Learning

PUBLICATIONS

Probabilistic Machine Learning: New Frontiers for Modeling Consumers and their Choices
Ryan Dew, Nicolas Padilla, **Lan E. Luo**, Shin Oblander, Asim Ansari, Khaled Boughanmi,
Michael Braun, Fred Feinberg, Jia Liu, Thomas Otter, Longxiu Tian, Yixin Wang, and
Mingzhang Yin¹
International Journal of Research in Marketing, Forthcoming

Public attitudes value interpretability but prioritize accuracy in Artificial Intelligence
Anne-Marie Nussberger, **Lan Luo**, L. Elisa Celis, and Molly J. Crockett
Nature Communications, 2022

WORKING PAPERS

How Visual Designs Drive Success: Interpretable Generative AI for Data-Driven Design
Lan E. Luo
Job Market Paper

Quantifying Discrimination Based on Facial Femininity: Using Controllable Stimuli Generation
for Hypothesis Testing
Lan E. Luo and Olivier Toubia
Invited for 3rd Round Review at Marketing Science (Minor Revision)

¹ From title footnote: "This paper is based on a session from the 2023 Triennial Choice Symposium. Ryan Dew and Nicolas Padilla organized the session. **Lan Luo** and Shin Oblander led the writing and editing process. All other authors contributed equally and are listed alphabetically."

SELECTED RESEARCH IN PROGRESS

“*” indicates equal authorship

To Read or Not To Read: How Individual Review Interactions Influence Search and Purchase
Lan E. Luo*, Sanjana Rosario*, Oded Netzer, and Verena Schoenmueller

Semantic Query Theory: Evidence from Retirement Benefit Claiming
Daniel Russman, **Lan E. Luo**, Alisa Wu, and Eric J. Johnson

A Timing-based Intervention for Divergent Delivery Bias in Meta Advertising A/B Tests
Lan E. Luo*, Dante Donati*, Oded Netzer*

Improving the Identifiability of Mechanistic Interpretability Methods with Causal Representation Learning
Lan E. Luo, Yixin Wang

AWARDS AND HONORS

AMA-Sheth Foundation Doctoral Consortium Fellow	2025
Columbia Marketing Division Research Grant	2024
ISMS Doctoral Consortium Fellow	2024
Columbia Experimental Laboratory in the Social Sciences Travel Grant	2024
Twelfth Triennial Choice Symposium Invitee	2023
W. Edwards Deming Center Doctoral Fellowship	2023
Bernstein Center for Leadership and Ethics Doctoral Research Grant	2022
Don Lehmann Scholar Award	2020
Yale Hartigan Undergraduate Senior Thesis Prize for Statistics & Data Science	2020
Richter Summer Fellowship Research Award	2019
Herb Scarf Fellowship Research Award in Economics	2018
Tobin Research Award in Economics	2018

CONFERENCE PRESENTATIONS

ISMS Marketing Science <i>How Visual Designs Drive Success: Interpretable Generative AI for Data-Driven Design</i>	2025
ISMS Marketing Science <i>Quantifying Discrimination Based on Facial Femininity: Using Controllable Stimuli Generation for Hypothesis Testing</i>	2024
Symposium on Artificial Intelligence in Marketing (Wisconsin School of Business) <i>Quantifying Discrimination Based on Facial Femininity: Using Controllable Stimuli Generation for Hypothesis Testing</i>	2024
12th Triennial Invitational Choice Symposium <i>Quantifying Discrimination Based on Facial Femininity: Using Controllable Stimuli Generation for Hypothesis Testing</i>	2023
ISMS Marketing Science <i>Quantifying Discrimination Based on Facial Femininity: Using Controllable Stimuli Generation for Hypothesis Testing</i>	2023

DOCTORAL COURSEWORK

Marketing:

Applied Multivariate Statistics
Consumer Behavior I
Analytical Models
Empirical Models
Bayesian Modeling and Computation
Mathematical Models
Consumer Behavior II
Marketing Decisions and Methods

Kamel Jedidi (F'20)
Michel Pham & Bernd Schmitt (F'20)
Kinshuk Jerath (S'21)
Asim Ansari (S'21)
Asim Ansari (F'21)
Rajeev Kohli (S'22)
Eric J. Johnson (S'22)
Donald Lehmann (Summer '22)

Economics:

Econometrics I
Economic Analysis I/II
Econometrics II
Economic Theory III/IV
Industrial Organization
Microeconometrics

José Luis Montiel Olea & Jushan Bai (F'20)
Geoffrey Heal (F'20)
Jushan Bai & Simon Lee (S'21)
Paolo Siconolfi (S'21)
Gautam Gowrisankaran (F'21)
Simon Lee (S'22)

Statistics and Computer Science:

Representation Learning
Probabilistic Models and Machine Learning
Applied Causality
Field Experiments, Machine Learning, and Causality

Carl Vondrick (F'21)
David Blei (F'22)
David Blei (S'23)
Donald Green & David Blei (S'24)

TEACHING

Teaching Assistant, Columbia Business School

Generative AI for Business (MBA)
Marketing Core (MBA)
Market Intelligence (MBA, MS)
Pricing Strategies (MBA)
Social and Economic Networks (MS)
Python Programming for Data Science (MS)

Olivier Toubia & Malek Ben Sliman (F'23)
Asim Ansari (F'24)
Kamel Jedidi & Robert Morais (F'21-24)
Asim Ansari (S'22-24)
Rajeev Kohli (F'22)
Yegor Tkachenko (Summer '23-24)

SKILLS

Language: English (native); Mandarin, French (proficient); Korean, Japanese (basic)

Programming: Python, R (fluent); STATA, Stan, Mathematica, Java, SQL (familiar)

REFERENCES

Olivier Toubia (*Advisor*)

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Eric Johnson (*Collaborator*)

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Oded Netzer (*Committee Member*)

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Hortense Fong (*Colleague*)

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SELECTED ABSTRACTS

How Visual Designs Drive Success: Interpretable Generative AI for Data-Driven Design

Lan E. Luo

Job Market Paper; Draft available upon request

Visual designs are often used in marketing (e.g., packaging, ads, media covers) to achieve a variety of business outcomes, like improved sales, click-through rates, and brand attitudes. Since designs are complex, unstructured data, it is difficult to determine what features drive their success in a way that is interpretable and managerially actionable. To address this challenge, I develop a novel methodological framework to automatically discover what interpretable features make visual designs in a given domain successful. I first leverage a deep generative text-to-image AI model (by fine-tuning Stable Diffusion 3.5 in my application) that adopts the role of designer and enables visual designs to be described by low-dimensional design representations. Then, I apply a novel adaptation of cutting-edge “mechanistic interpretability” methods—specifically “sparse autoencoders” typically applied to large language models—to scalably discover a taxonomy of interpretable and managerially relevant features predictive of success from these design representations. Finally, I generate image redesigns by manipulating features of interest to help managers scalably pilot data-driven design changes. I apply this framework to discover how book cover redesigns predict sales on Amazon.com using a unique dataset I collected of over 160,000 books. I discover a diverse set of interpretable features related to illustration, typography, composition, and layout. I then create realistic cover redesigns predicted to improve sales by manipulating those features (e.g., redesigns with lower contrast and less separation of text and graphical elements). In a holdout analysis with a rich set of control variables, including just 30 of these discovered features (out of 9,728) improves variation explained in sales by nearly as much as prices and by more than reviews. Back-of-the-envelope calculations suggest that a large publisher could leverage this subset of features to increase annual revenue by over \$9.1 million, reflecting a change in sales equivalent to introducing an 8.5% price discount.

Quantifying Discrimination Based on Facial Femininity: Using Controllable Stimuli Generation for Hypothesis Testing

Lan E. Luo and Olivier Toubia

Working Paper; Invited for 3rd Round Review at Marketing Science (Minor Revision)

We document an understudied form of gender bias based on facial femininity. To do so, we use a Generative Adversarial Network to create realistic photos of people and then manipulate their facial femininity independently of other attributes like pose, general facial expression, background, hairstyle, and clothes. Then, we devise an experimental design that allows us to identify the separate and combined causal impact of facial femininity and gender identity (proxied by gender pronouns) on real-

life outcomes. In a field experiment within a naturalistic advertising environment, we find that prospective customers of an education company discriminated against facial femininity at various stages of the purchase funnel, to a similar extent for men, women, and non-binary people. The findings inform managers and policymakers of an important form of discrimination based on faces. Methodologically, we introduce a novel framework for testing specific hypotheses pertaining to causal effects of treatments which are derived from unstructured data. The approach involves using natural text to readily identify hypothesis-relevant features from a generative AI model (in a particularly disentangled representation space) and then creating realistic stimuli that vary controllably in those features.

Probabilistic Machine Learning: New Frontiers for Modeling Consumers and their Choices

Ryan Dew, Nicolas Padilla, **Lan E. Luo**, Shin Oblander, Asim Ansari, Khaled Boughanmi, Michael Braun, Fred Feinberg, Jia Liu, Thomas Otter, Longxiu Tian, Yixin Wang, and Mingzhang Yin
International Journal of Research in Marketing, Forthcoming

Making sense of massive, individual-level data is challenging: marketing researchers and analysts need flexible models that can accommodate rich patterns of heterogeneity and dynamics, work with and link diverse data types, and scale to modern data sizes. Practitioners also need tools that can quantify uncertainty in models and predictions of consumer behavior to inform optimal decision-making. In this paper, we demonstrate the promise of probabilistic machine learning (PML), which refers to the pairing of probabilistic modeling and machine learning methods, in pushing the frontier of combining flexibility, scalability, interpretability, and uncertainty quantification for building better models of consumers and their choices. Specifically, we overview both PML models and inference methods, and highlight their utility for addressing four common classes of marketing problems: (1) uncovering heterogeneity, (2) flexibly modeling nonlinearities and dynamics, (3) handling high-dimensional and unstructured data, and (4) addressing missingness, often via data fusion. We also discuss promising directions in enriching marketing models, reflecting recent developments in representation learning, causal inference, experimentation and decision-making, and theory-based behavioral modeling.

Public attitudes value interpretability but prioritize accuracy in Artificial Intelligence

Anne-Marie Nussberger, **Lan Luo**, L. Elisa Celis, and Molly J. Crockett
Nature Communications, 2022

As Artificial Intelligence (AI) proliferates across important social institutions, many of the most powerful AI systems available are difficult to interpret for end-users and engineers alike. Here, we sought to characterize public attitudes towards AI interpretability. Across seven studies ($N = 2475$), we demonstrate robust and positive attitudes towards interpretable AI among non-experts that generalize

across a variety of real-world applications and follow predictable patterns. Participants value interpretability positively across different levels of AI autonomy and accuracy, and rate interpretability as more important for AI decisions involving high stakes and scarce resources. Crucially, when AI interpretability trades off against AI accuracy, participants prioritize accuracy over interpretability under the same conditions driving positive attitudes towards interpretability in the first place: amidst high stakes and scarce resources. These attitudes could drive a proliferation of AI systems making high-impact ethical decisions that are difficult to explain and understand.

To Read or Not To Read: How Individual Review Interactions Influence Search and Purchase

Lan E. Luo*, Sanjana Rosario*, Oded Netzer, and Verena Schoenmueller

Work in Progress

While extant marketing literature establishes the impact of individual reviews on purchase (Liu et al. 2019; Vana and Lambrecht 2021) and search (Varga and Albuquerque 2024), less is known about when and how consumers engage with individual reviews as well as what information they seek. Motivated by this gap, we study user behavior on a large e-commerce platform with over 700,000 monthly product impressions and detailed logs of review-reading and purchase decisions. We exploit plausibly exogenous variation in the displayed set of individual reviews for a product and causally establish two novel empirical findings. First, we find evidence that discrepancies between an individual review's star rating and the holistic evaluation of that review (reflecting both its text and star rating) influence purchase, an effect that is driven partly by reference dependence (to the star rating) and negativity bias. To measure discrepancies, we develop a novel large language encoder model architecture (based on DeBERTaV3) that takes as input an individual review's text and star rating to predict a holistic, continuous sentiment score for that review. We find that a two-star review that reads overall like a three-star review (which may elicit pleasant surprise) increases purchase likelihood by 2.8% compared to a four-star review that reads overall like a three-star review (which may elicit unpleasant disappointment). Second, we find evidence that users are more likely to engage with reviews when product quality is weakly signaled. To measure signal strength, we compute the semantic variance across review and product sentence embeddings and apply transformer-based classifiers to detect when reviews are undermined by conflicting information, disagreement, and ambivalent language. This effect is significant only for highly rated products, suggesting that users may have less motivation to resolve diverging reviews for lower-rated products.

Semantic Query Theory: Evidence from Retirement Benefit Claiming

Daniel Russman, **Lan E. Luo**, Alisa Wu, and Eric J. Johnson

Working Paper

Citizens in many countries need to decide when to claim retirement benefits. That usually presents a tradeoff between claiming earlier for a smaller monthly payment or waiting for a larger payment. Although there is no single right decision for everyone, prior analyses of Social Security claiming data suggests that Americans, on average, claim too early, leaving significant amounts of money on the table. To better understand and guide this important decision, we first use natural language processing methods to uncover the semantic themes (i.e., topics) considered in making retirement benefit claiming decisions. We then build on existing theories of memory-based preferences to explain and influence consumers' decision-making processes. Specifically, we propose Semantic Query Theory as a novel theoretical framework to help both explain decision processes and design interventions to improve outcomes. Across four studies, we discover and validate semantic themes present when consumers consider their decision to claim retirement benefits using methods based on probabilistic machine learning (e.g., structural topic modeling) and large language modeling; estimate the associative network of such topics in memory using discrete-time, discrete-state Markov processes; and leverage this information in an intervention that delays benefits claiming by 8.9 months, on average.